M1: Specification, Wireframes and Accessibility Plan

Student Finance Tracker · Summative Assignment: Building Responsive UI

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Repository: github.com/David-Harold/student-finance-tracker
Live site: david-harold.github.io/student-finance-tracker/

1. Purpose and decisions

A vanilla HTML, CSS and JavaScript tracker for a student who budgets a monthly allowance and spends in more than one currency. The core problem it addresses: money running out before the month ends.

Topic	Decision
Theme	Student Finance Tracker
Primary user	A student in Kigali who budgets a monthly allowance and spends in RWF, USD and GBP
Currencies	Base RWF; USD and GBP as the other two. Rates are entered manually in Settings as RWF per one unit. No API.
Records layout	Cards below 768 px; a semantic table from 768 px upward
Charts	One chart: last-7-days spending, drawn with CSS bars, with a text alternative. It is separate from the records layout.
Regex	Four required validation rules plus two advanced rules: back-reference (repeated word) and lookahead (non-zero rate or cap)
Distinguishing feature (proposed)	**Month-end runway**: from the monthly cap and the spending so far, show a safe daily spend and a projection of the day the cap would run out. Pending the author's confirmation.
Animation	One transition (the cap bar easing to its new width), disabled under prefers-reduced-motion
Out of scope	Stretch items (service worker, theme toggle, CSV export, jQuery page)
Open decision (M3)	Repeated-word rule: block the form, or warn and allow. Decide when writing the validators.

2. Data model

Record (stored as an array). The currency field is an addition to the brief's model, required so mixed-currency spending can be totalled.

Field	Type	Rule
id	string	`rec_0001` format. Next id = highest numeric suffix + 1, zero-padded to 4 digits (never array length, to avoid duplicates after deletes).
description	string	Trimmed, internal whitespace collapsed; rules R1 and R5
amount	number	Rule R2: no sign, no thousands separator, at most two decimals
currency	string	`RWF`, `USD` or `GBP`
category	string	Rule R4; defaults Food, Books, Transport, Entertainment, Fees, Other (editable)
date	string	`YYYY-MM-DD`; rule R3 plus a real-calendar check
createdAt / updatedAt	string	ISO 8601. createdAt set once; updatedAt refreshed on every edit.

Settings (separate storage key)

Field	Type	Rule
schemaVersion	number	1; allows later migration
baseCurrency	string	`RWF`
rates	object	`{USD, GBP}`: RWF per one unit; rule R6 (positive, lookahead)
monthlyCap	number	In base currency; rule R6
categories	array of string	Each name validated by R4; deleting a category in use reassigns its records to Other

Item	Definition
Storage keys	`sft:records`, `sft:settings`
Converted amount	baseAmount = amount × rate(currency); rate(RWF) = 1
Month spend	Sum of baseAmount for records dated in the current calendar month
Remaining / overage	monthlyCap – month spend (negative means overage)
Safe daily spend	remaining ÷ days left in month (today included); shown only while remaining > 0
Projection	average daily spend = month spend ÷ day of month; cap reached on day ceil(cap ÷ average). Message only if that day falls before month end.
Display rounding	RWF whole numbers; USD and GBP two decimals; sums computed before rounding; Intl.NumberFormat for output

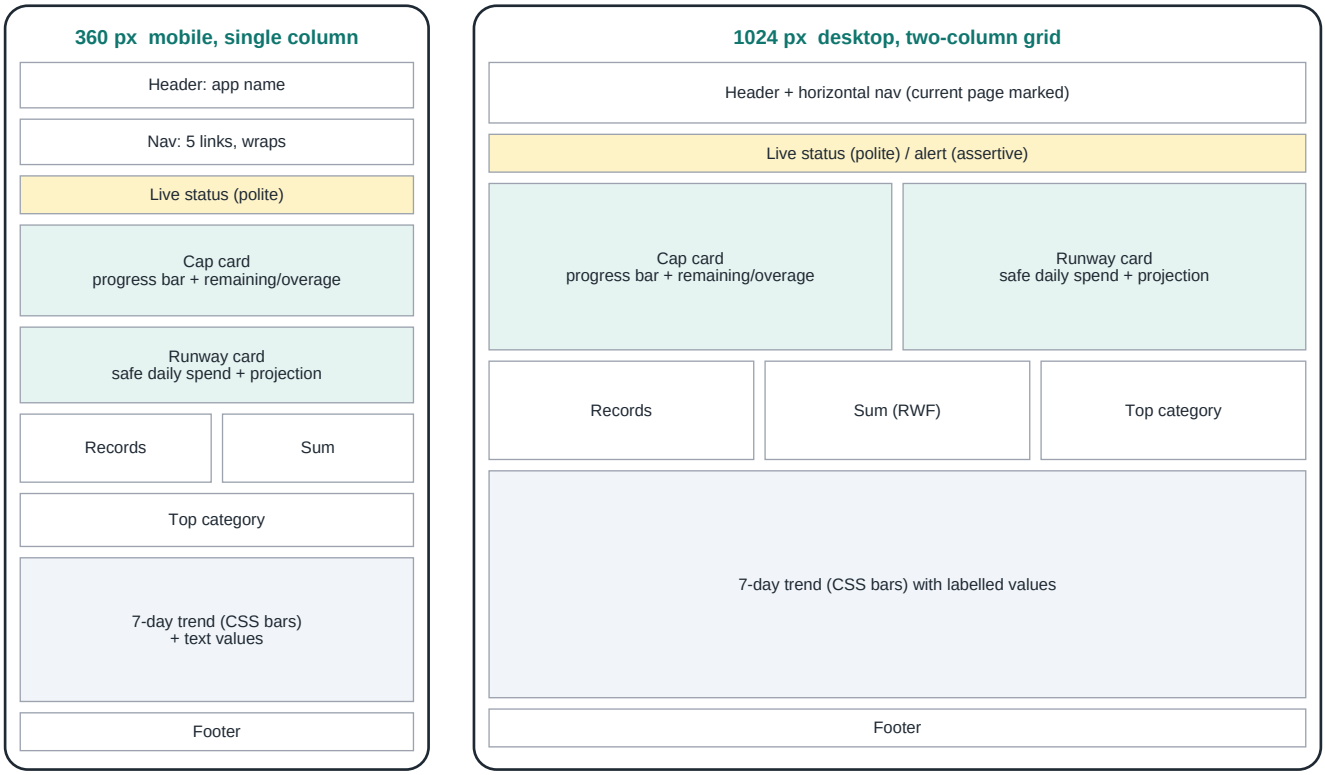
Sample record

```
{ "id": "rec_0001", "description": "Lunch at cafeteria", "amount": 3500, "currency": "RWF", "category": "Food", "date": "2026-10-09", "createdAt": "2026-10-09T09:12:00.000Z", "updatedAt": "2026-10-09T09:12:00.000Z" }
```

3. Wireframes and page content

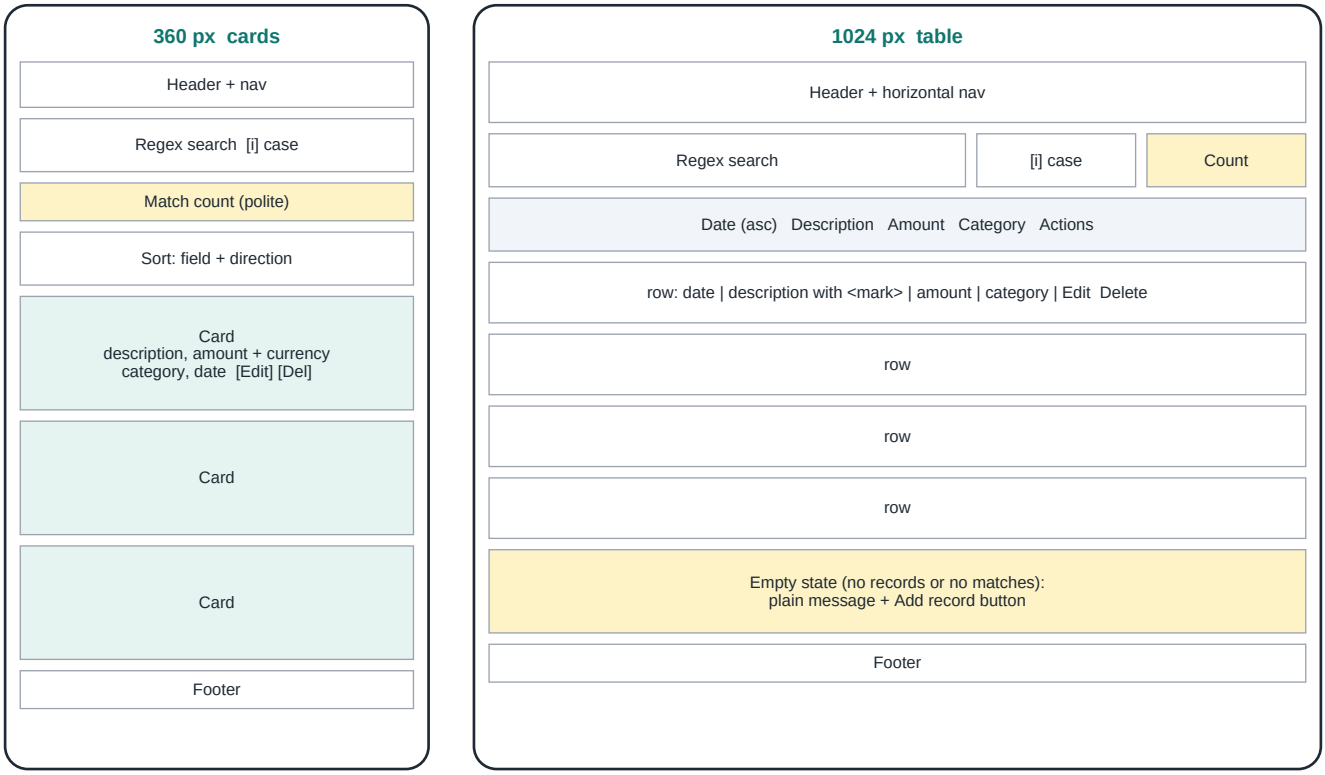
Schematic drafts from the agreed decisions. Shaded green: cards. Amber: live regions. Blue-grey: chart. The 768 px tablet layout sits between the two shown (two-column dashboard, table appears).

Figure 1. Dashboard



Skip link (not drawn) is the first focusable element and becomes visible on focus.

Figure 2. Records



Sorting by date, description and amount; matches highlighted with the mark element.

Section	Content
About	Purpose of the tracker; author contact (GitHub profile and email); short note on how to use it
Dashboard	Cap card, runway card, record count, sum in RWF, top category (by spend, ties listed), 7-day chart, live regions
Records	Regex search with case toggle, sort controls, cards or table, inline edit, confirmed delete, empty and no-match states
Add / Edit	Description, amount, currency, category, date; inline messages beside each field; edit reuses the same validators
Settings	Base currency (fixed RWF), USD and GBP rates, monthly cap, category list, export and import JSON, import report

4. Accessibility plan

Area	Plan	Verified by
Landmarks	header, nav (labelled), main, one section per page area, footer; one h1, then h2 and h3 in order	Heading outline and landmark list in the browser accessibility tree
Skip link	First focusable element, visible on focus, targets main	Keyboard pass
Forms	Visible label bound with for/id; error text linked with aria-describedby; aria-invalid while failing; errors never colour-only	Keyboard and screen-reader spot check
Live regions	Present and empty in the initial HTML. Polite region: saved, count of matches, under-cap messages, import report. Assertive region: cap exceeded, import rejected.	Trigger each message and read its announcement
Keyboard	Tab order follows visual order; Enter or Space activates buttons; Esc cancels an inline edit or closes the delete dialog; focus returns to the triggering button	Full keyboard-only run: add, edit, delete, search, sort, export, import
Focus	Visible :focus-visible outline on every control, never removed without replacement	Keyboard pass at all three widths
Table	caption, th scope=col, aria-sort on the sorted column header	Inspect markup
Highlights	mark element with AA-contrast colours; matches added through DOM text nodes, never as an HTML string	Search a description containing angle brackets
Chart	Bars carry visible value labels and the same data is available as text	Inspect without CSS
Motion	One transition; removed under prefers-reduced-motion	Toggle the OS setting
Touch targets	About 44 px minimum on controls	Inspect at 360 px

Colour contrast (WCAG 2.x ratios computed from the palette)

Pair	Foreground	Background	Ratio	Needs	Result
Body text	`#1f2933`	`#ffffff`	14.76:1	4.5:1	Pass
Muted text	`#52606d`	`#ffffff`	6.46:1	4.5:1	Pass
Link and accent text	`#0f766e`	`#ffffff`	5.47:1	4.5:1	Pass
Button label on accent	`#ffffff`	`#0f766e`	5.47:1	4.5:1	Pass
Error text	`#b91c1c`	`#ffffff`	6.47:1	4.5:1	Pass
Error text on error panel	`#991b1b`	`#fee2e2`	6.80:1	4.5:1	Pass
Warning text on warning panel	`#92400e`	`#fef3c7`	6.37:1	4.5:1	Pass
Highlight (mark) text	`#1f2933`	`#fde68a`	11.85:1	4.5:1	Pass
Focus outline vs page (UI, 3:1)	`#1d4ed8`	`#ffffff`	6.70:1	3.0:1	Pass

5. Regex catalog (planned)

ID	Field	Pattern	Valid	Invalid	Message
R1	Description	`^\S(?:.*\S)?$`	Lunch at cafeteria	" Lunch"	No leading or trailing spaces.
R2	Amount	`^(0|[1-9]\d*)(\.\d{1,2})?$`	12.50, 0.5, 100	.75, 12., 01.5	Use digits with up to two decimals, e.g. 12.50.
R3	Date	`^\d{4}-(0[1-9]|1[0-2])-(0[1-9]|[12]\d|3[01])$`	2024-02-29	2025-9-29, 2025-13-01	Use YYYY-MM-DD. A second step rejects impossible days (2025-02-31).
R4	Category	`^[A-Za-z]+(?:[ -][A-Za-z]+)*$`	Mobile Money	Food2, Food-	Letters, spaces and hyphens only.
R5	Description (advanced, back-reference)	`\b(\w+)\s+\1\b`, flag i	Coffee with friends	the the lunch	Repeated word. Block or warn: open decision.
R6	Rates and cap (advanced, lookahead)	`^(?=.*[1-9])\d+(?:\.\d+)?$`	1450, 0.05	0, 0.00	Enter a number greater than zero.

- **Normalisation:** runs of whitespace in the description are collapsed to a single space before R1 is applied.
- **Search:** pattern compiled inside try/catch; invalid patterns show a calm message and match nothing; case toggle switches the i flag; pattern length capped to limit backtracking; no global flag on the filtering regex.
- **Search examples to document:** `\.\d{2}\b` (cents present), `(coffee|tea)` with i (beverages), `\b(\w+)\s+\1\b` (duplicate word).

6. Modules, tests and commits

File	Responsibility	Must not
`scripts/validators.js`	R1 to R6, real-date check, whitespace normalisation	Touch the DOM
`scripts/storage.js`	Load, save, import and export with structure validation	Touch the DOM
`scripts/state.js`	Records and settings, add, update, delete, ids and timestamps, subscribers	Read the DOM
`scripts/search.js`	Safe regex compiler, filtering, match segments for highlighting	Write HTML strings
`scripts/stats.js`	Totals, top category, 7-day buckets, cap, runway	Know about presentation
`scripts/ui.js`	Render cards, table, dashboard, forms; announcements; events	Hold business rules
`scripts/main.js`	Wire the modules together	Contain logic

Repository layout: `index.html, tests.html, seed.json, README.md, .gitignore, styles/, scripts/, assets/`. Lowercase file names throughout. Flow: event, validate, change state, save, re-render.

tests.html

At least two valid and two invalid inputs per rule, with boundaries (0, 0.00, 007, .5, 12.345, padded text); the real-date check (2024-02-29 passes, 2025-02-31 fails); the search compiler returning nothing for an invalid pattern; the runway formulas for a known month. Output as pass or fail text, not colour alone.

seed.json

At least ten records: leap day, year boundary, month ends, the current month for the chart and runway, amounts of 0.01 and a large value with cents, all three currencies, all six categories, a single-character description, a long description, and text with quotes, ampersands and angle brackets. Every record passes the app's own

validators.

Commit plan

Milestone	Commit message
M1	M1: add specification, wireframes and accessibility plan
M2	M2: add semantic HTML structure and mobile-first base CSS
M3	M3: add regex validators, inline errors and tests.html
M4	M4: add rendering, sorting and safe regex search
M5	M5: add dashboard stats, cap and runway with live regions
M6	M6: add persistence, JSON import and export, settings
M7	M7: polish, accessibility audit, README and demo notes

7. Timetable to the deadline

Deadline: Wednesday 14 October 2026, 11:59 pm. Submit on the official page only, targeting about 9 pm.

When	Work	Exit test
Tue 13 Oct, morning	M1 commit; M2 skeleton and base CSS; .gitignore; first Pages build of index.html	Skeleton loads at the live URL
Tue 13 Oct, afternoon	M3: validators, inline errors, tests.html	All assertions pass
Tue 13 Oct, evening	M4: render, sort, regex search	Invalid pattern does not break the page
Wed 14 Oct, morning	M5 and M6: stats, cap, runway, persistence, import and export, settings	Export, clear, import gives identical data
Wed 14 Oct, afternoon	M7: keyboard pass, contrast check, README, demo video, deployed-site check	Keyboard-only run succeeds on the live site
Wed 14 Oct, by 9 pm	Final live-URL check in a private window; submit	Submitted

> **Scope rule**
>
> If time runs short: keep the required features working and polished; drop the runway projection first, then animation; never drop import validation, keyboard access or the README.

Documentation drafted with AI assistance, which the brief permits for documentation. All application code is written by the author.